Kubernetes Deployment Assignment

This project demonstrates containerization and deployment of a Node.js application on Kubernetes. The application consists of a frontend UI and backend API. Two versions of the application were created to demonstrate rolling updates.

2. Application Overview

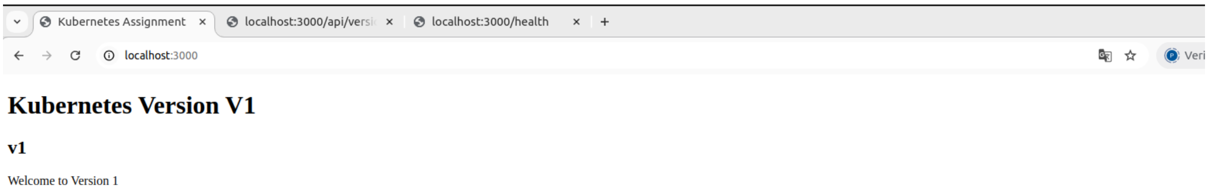
Frontend

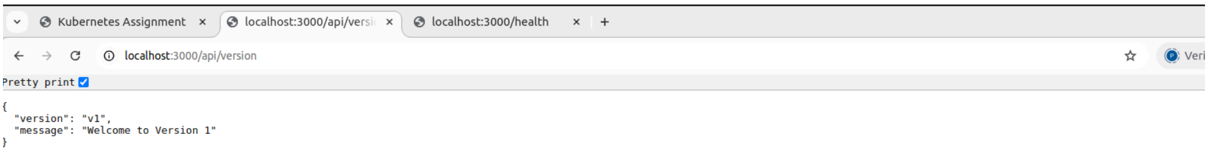
Displays application version information retrieved from the backend API.

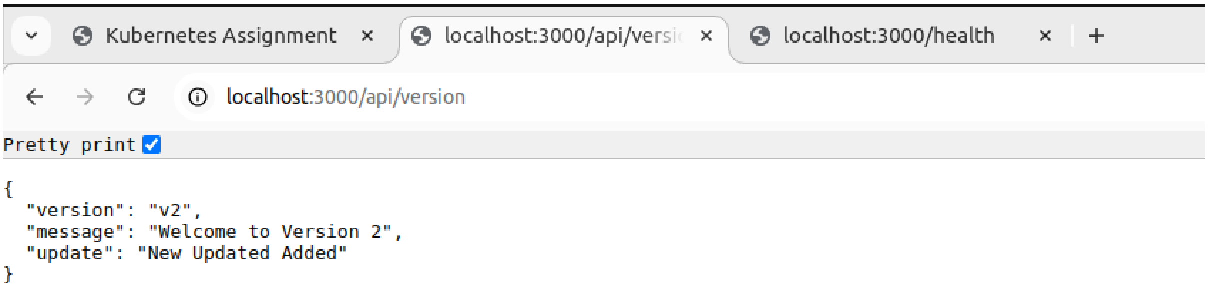
Backend

Node.js + Express application exposing:

GET /api/version
GET /health







3. Docker Image Creation

Dockerfile ExplanatioN

Base Image: node:22-alpine
Working Directory: /app
Dependencies installed using npm install
Application exposed on port 3000

Version 1 Image

Commands:

docker build -t kubernetes_ass:v1 .
docker run -d -p 3000:3000 kubernetes_ass:v1

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/v2$ docker images
REPOSITORY                                TAG         IMAGE ID       CREATED           SIZE
kubernetes_ass                            v2          5680e984bb51   5 seconds ago     171MB
kubernetes_ass                            v1          cfcad9689485   13 minutes ago    169MB
```

4.Kubernetes POD

Commands used to view pods:

kubectl get pods
kubectl describe pod k8s-demo-pod

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl apply -f pod.yml
pod/k8s-demo-pod created
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods
NAME           READY   STATUS    RESTARTS   AGE
k8s-demo-pod   1/1     Running   0          10s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl describe pod assignment-pod
Error from server (NotFound): pods "assignment-pod" not found
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl describe pod k8s-demo-pod
Name:             k8s-demo-pod
Namespace:        default
Priority:         0
Service Account:  default
Node:             minikube/192.168.39.14
Start Time:       Sat, 13 Jun 2026 17:26:57 +0530
Labels:           app=k8s-demo
Annotations:      <none>
Status:           Running
IP:               10.244.0.5
IPs:
  IP:  10.244.0.5
Containers:
  k8s-demo-container:
    Container ID:   docker://7d23aa99cf053baf037ea8edd038b0af37d7d277045ef339c36719d0494bc3ca
    Image:          kubernetes_ass:v2
    Image ID:       docker://sha256:aae64a7873a512d0aa65f308b32e9e99a131306742f8191cd071949cf0121ce0
    Port:           3000/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Sat, 13 Jun 2026 17:26:57 +0530
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-88lns (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
Volumes:
  kube-api-access-88lns:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
    ConfigMapOptional:       <nil>
    DownwardAPI:             true
QoS Class:                   BestEffort
Node-Selectors:              <none>
Tolerations:                 node.kubernetes.io/not-ready:NoExecute op=Exists for 300s
                             node.kubernetes.io/unreachable:NoExecute op=Exists for 300s
Events:
```

5. Kubernetes ReplicaSet

kubectl get rs

kubectl get pods

kubectl delete pod

```
No resources found in default namespace.
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl apply -f rs.yml
replicaset.apps/k8s-demo-rs created
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get rs
NAME          DESIRED   CURRENT   READY   AGE
k8s-demo-rs   3         3         3       16s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
k8s-demo-rs-7fncc   1/1     Running   0          20s
k8s-demo-rs-j74zf   1/1     Running   0          20s
k8s-demo-rs-m8hfr   1/1     Running   0          20s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl delete pod k8s-demo-rs-7fncc
pod "k8s-demo-rs-7fncc" deleted
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods -w
NAME                READY   STATUS    RESTARTS   AGE
k8s-demo-rs-j74zf   1/1     Running   0          97s
k8s-demo-rs-jgkpt   1/1     Running   0          39s
k8s-demo-rs-m8hfr   1/1     Running   0          97s
```

6. Kubernetes Deployment

kubectl get deployments
kubectl get pods

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl apply -f deployment.yml
deployment.apps/k8s-demo-deployment created
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
k8s-demo-deployment-7d76597f5-blj6l    1/1     Running   0          12s
k8s-demo-deployment-7d76597f5-cf2rg    1/1     Running   0          12s
k8s-demo-deployment-7d76597f5-lwpc2    1/1     Running   0          12s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get rs
NAME                             DESIRED   CURRENT   READY   AGE
k8s-demo-deployment-7d76597f5    3         3         3       19s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get deployments
NAME                  READY   UP-TO-DATE   AVAILABLE   AGE
k8s-demo-deployment   3/3     3            3           27s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$
```

7. Health Probes

Liveness Probe

Detects unhealthy containers and restarts them.

Readiness Probe

Determines if the application is ready to receive traffic.

Startup Probe

Allows slow-starting applications time to initialize.

8. Service Exposure

kubectl get svc

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ vi service.yml
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl apply -f service.yml
service/k8s-demo-service created
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get svc
NAME               TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
k8s-demo-service   NodePort    10.111.187.102   <none>        80:31531/TCP   7s
kubernetes         ClusterIP   10.96.0.1        <none>        443/TCP        496d
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ minikube service k8s-demo-service
|-----------|------------------|-------------|---------------------------|
| NAMESPACE |       NAME       | TARGET PORT |            URL            |
|-----------|------------------|-------------|---------------------------|
| default   | k8s-demo-service |          80 | http://192.168.39.14:31531 |
|-----------|------------------|-------------|---------------------------|
🎉  Opening service default/k8s-demo-service in default browser...
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ Opening in existing browser session.
```

9. Rolling Update Demonstration

Initial State

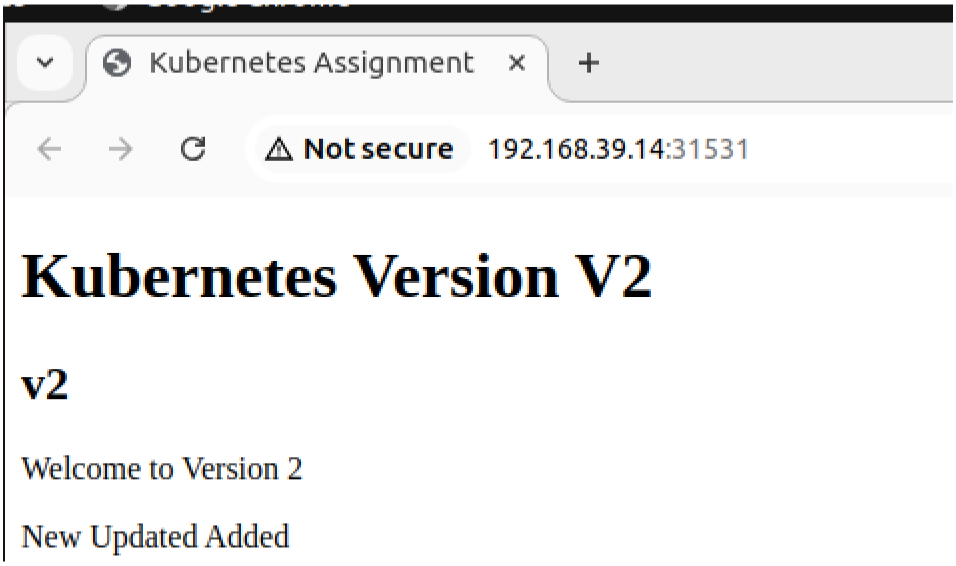
```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods -w
NAME                                  READY   STATUS        RESTARTS   AGE
k8s-demo-deployment-7d76597f5-blj6l   1/1     Terminating   0          5m5s
k8s-demo-deployment-7d76597f5-cf2rg   1/1     Terminating   0          5m5s
k8s-demo-deployment-7d76597f5-lwpc2   1/1     Terminating   0          5m5s
k8s-demo-deployment-998fb79b4-42w8q   1/1     Running       0          29s
k8s-demo-deployment-998fb79b4-4g6t9   1/1     Running       0          24s
k8s-demo-deployment-998fb79b4-799n5   1/1     Running       0          29s
k8s-demo-deployment-7d76597f5-cf2rg   0/1     Error         0          5m6s
k8s-demo-deployment-7d76597f5-cf2rg   0/1     Error         0          5m6s
k8s-demo-deployment-7d76597f5-cf2rg   0/1     Error         0          5m6s
k8s-demo-deployment-7d76597f5-lwpc2   0/1     Error         0          5m11s
k8s-demo-deployment-7d76597f5-lwpc2   0/1     Error         0          5m12s
k8s-demo-deployment-7d76597f5-lwpc2   0/1     Error         0          5m12s
k8s-demo-deployment-7d76597f5-blj6l   0/1     Error         0          5m12s
k8s-demo-deployment-7d76597f5-blj6l   0/1     Error         0          5m13s
k8s-demo-deployment-7d76597f5-blj6l   0/1     Error         0          5m13s
```

kubectl set image deployment/k8s-demo-deployment

\

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl rollout status deployment/k8s-demo-deployment
deployment "k8s-demo-deployment" successfully rolled out
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$
```

```
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl describe deployment k8s-demo-deployment | grep Image
    Image:          kubernetes_ass:v1
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl set image deployment/k8s-demo-deployment \
k8s-demo-container=kubernetes_ass:v2
deployment.apps/k8s-demo-deployment image updated
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl get pods
NAME                                   READY   STATUS    RESTARTS   AGE
k8s-demo-deployment-998fb79b4-42w8q    1/1     Running   0          74s
k8s-demo-deployment-998fb79b4-4g6t9    1/1     Running   0          69s
k8s-demo-deployment-998fb79b4-799n5    1/1     Running   0          74s
npci-admin@HYDD001:~/Desktop/AIML/kubs_Ass/k8s$ kubectl describe deployment k8s-demo-deployment | grep Image
    Image:          kubernetes_ass:v2
```



CONCLUSION:

This assignment demonstrated containerization using Docker and orchestration using Kubernetes. The application was deployed using Pods, ReplicaSets, and Deployments. Health probes were configured and a rolling update from Version 1 to Version 2 was successfully performed without downtime.